Bayesian Inference for the German Tank Problem on Real Data: Conjugate Models, MCMC and Hierarchical Analysis across Nine Open Datasets

Project Proposal — Version 3 (data-analysis focus)

BSc Statistical Data Science, Year 3
Module: Bayesian Inference

November 5, 2026

Abstract

The German Tank Problem estimates the size N of a population labelled $1, \ldots, N$ from a random subset of k labels. Its appeal as a project is that, for many real labelled populations, the true N is *known*, so Bayesian methods can be tested on genuine data rather than on simulations alone. This proposal attaches an open dataset, a concrete analysis protocol and a ground-truth check to every component of the Bayesian syllabus: exact posteriors and Bayes factors on e-commerce invoice numbers and CryptoPunks token IDs, a random-sample-size thinning model fitted by Gibbs and Metropolis–Hastings, the Jelinski–Moranda Gamma–Exponential model on classical software failure data, spike-and-slab regression and hierarchical models on open-source defect data, and a two-stage wildlife model on bird-banding and capture–recapture data. A semi-synthetic counterfeit experiment on real aircraft serial numbers closes the study. All data are open and links are given.

1 What changes in this version

Versions 1–2 described the models. This version adds the data. For each workstream it states the dataset (with link), what is the population and what is the sample, whether a ground truth exists, how the data will be cleaned, which analyses are run and what counts as success. Two principles organise the analysis:

- **Retrospective subsampling.** When the complete label set is known (invoices, token IDs, band blocks), draw many random subsets of size k, apply every method, and compare with the known N_{true}. This yields coverage, calibration and scoring-rule results on real labels.
- **Natural sampling.** When the sample arises from a real process (invoices with a recorded customer, tokens that were traded, aircraft still on the register), the sampling is *not* uniform in general. These cases test the selection-bias and thinning models and show what happens when the textbook assumption is violated.

2 Data inventory

A tenth item, the historical WWII table, gives context only (Section 5).

ID	Dataset	Population → sample	Ground truth
D1	Online Retail (UCI)	invoices → random or recorded-customer subset	yes: full invoice list
D2	Online Retail II (UCI)	weekly invoice flow, two years	yes: weekly counts
D3	Software failure data (Musa/DACS, via R package)	defects → failure times	partial: later failures held out
D4	PROMISE/Jureczko defect data; CRAN `hotspot`	classes with post-release bug counts and 20 code metrics	observed counts (no serial labels)
D5	CryptoPunks (Kaggle)	token IDs → traded tokens	yes: supply 10,000
D6	Eight NFT collections (Cornell eCommons)	several collections of known size	yes: supply per collection
D7	FAA Releasable Aircraft Registry	manufacturer serials → aircraft still registered	partial: published production totals
D8	USGS North American Bird Banding Program	band numbers → encountered birds	yes within a band block
D9	Rcapture `hare` (capture–recapture)	snowshoes hares, 6 occasions	published estimates ≈ 77–81

2.1 Sources and access

- **D1 Online Retail**, UCI Machine Learning Repository (CC BY 4.0). All transactions of a UK-based online retailer between 01/12/2010 and 09/12/2011, 541,909 rows; `InvoiceNo` is a six-digit number unique to each transaction, with a leading "C" for cancellations. https://archive.ics.uci.edu/dataset/352/online+retail. DOI: https://doi.org/10.24432/C5BW33. In Python: `from ucimlrepo import fetch_ucirepo; fetch_ucirepo(id=352)`. Many rows have no `CustomerID`, which gives a natural thinning mechanism.
- **D2 Online Retail II**, UCI. Two years of transactions of the same type, 1,067,371 rows. DOI: https://doi.org/10.24432/C5CG6D.
- **D3 Software failure data (DACS/Musa).** Inter-failure times in CPU seconds for several real systems; `sys1` has 136 failures and `ss3` has 278. Documented in the R package `Rsrat`: https://rdrr.io/github/okamumu/Rsrat/man/dacs.html. The data come from M. R. Lyu, *Handbook of Software Reliability Engineering* (1996), with book pages at http://www.cse.cuhk.edu.hk/~lyu/book/reliability/.
- **D4 PROMISE / Jureczko defect data.** Per-class post-release bug counts and 20 object-oriented code metrics for dozens of releases of open-source Java projects (Ant, Camel, Ivy, jEdit, Log4j, Lucene, POI, Synapse, Velocity, Xalan, Xerces and others). The original host http://openscience.us/repo/defect/ck/ is not always online, so use the CRAN package `hotspot`, which bundles per-class bug counts for 14 releases of Ant, jEdit, Log4j and Xerces: https://cloud.r-project.org/web/packages/hotspot/refman/hotspot.html. A further mirror is on IEEE DataPort: https://ieee-dataport.org/documents/software-defect (DOI 10.21227/H2K078; free account may be needed). The Eclipse bug data are described at https://www.st.cs.uni-saarland.de/softevo/bug-data/eclipse/.
- **D5 CryptoPunks**, Kaggle (B. Tunguz): all CryptoPunk transactions since 2017 and images of the 10,000 punks. https://www.kaggle.com/datasets/tunguz/cryptopunks. An alternative transaction file is `zeinalamah/cryptopunks-transactions` on Kaggle.
- **D6 Eight NFT collections**, Cornell eCommons: transaction history and visual traits for eight highly valued collections (including Bored Ape Yacht Club) up to 31 March 2022,

from the paper *Non-fungible token transactions: data and challenges*. https://ecommons.cornell.edu/entities/publication/ca3b44d5-0241-4a15-a8f0-7e44c7572e62.

- **D7 FAA Releasable Aircraft Database** (public domain). One row per registered aircraft with the manufacturer-assigned serial number (free text, up to 30 characters), model code and year, plus a reference table of makes and models. Download: https://registry.faa.gov/database/ReleasableAircraft.zip; layout: https://registry.faa.gov/database/ardata.pdf.
- **D8 North American Bird Banding Program (USGS BBL)**, CC0. More than 84 million banding, encounter and recapture records from 1960. Each federal band carries a unique nine-digit number. ScienceBase release with zipped files of several hundred MB each: https://sciencebase.gov/catalog/item/68837a85d4be027deba86316 (DOI 10.5066/P1KPZGAR). Smaller extracts can be requested through the Bander Portal: https://www.usgs.gov/labs/birdb-lab/science/banding-and-encounter-data-requests.
- **D9 Rcapture data.** `hare` (68 snowshoe hares, six capture occasions), with published abundance estimates of about 77 to 81. https://search.r-project.org/CRAN/refmans/Rcapture/html/hare.html; package: https://cran.r-universe.dev/Rcapture.
- **Historical benchmark.** The WWII table (statistical estimates 169, 244, 327 versus German records 122, 271, 342): https://en.wikipedia.org/wiki/German_tank_problem. For the multiple-factory variant see https://arxiv.org/pdf/2403.14881v4.

No open dataset of *labelled* counterfeit serial numbers was found, so the counterfeit study (WP7) injects contamination at a known rate into real serial series and says so explicitly.

3 Syllabus to data map

Syllabus topic	**Data**	**Concrete analysis**
Conjugate priors and exponential families	D1, D2, D3, D9	Beta–Bernoulli and Gamma–Poisson thinning of invoices (D1); Normal–Gamma on weekly log-volumes (D2); Gamma–Exponential on failure times (D3); Beta–Binomial recapture (D9); Dirichlet–Multinomial shares across aircraft models (D7)
Bayesian estimation	D1, D5	Posterior mean, median, MAP, HPD for N over thousands of real-label subsamples; coverage and log score against known N_{true}
Hypothesis testing	D1, D5, D7	Bayes factors for "$N \leq L$" with L below and above the truth; false-alarm and power curves; Savage–Dickey test for selection bias
Choice of priors	D1, D5	Flat, $1/n$, truncated, Poisson and negative binomial priors, with hyperparameters set deliberately wrong; sensitivity of intervals and Bayes factors
MCMC (Gibbs, MH, diagnostics)	D1, D2, D5	Gibbs for thinning (D1); Metropolis–Hastings for selection bias (D1, D5); trace, ACF, $\hat{R}$, ESS; comparison with exact grid posterior
Bayesian linear models	D2, D4	Normal-Inverse-Gamma regression, ridge and LASSO as MAP, spike-and-slab selection among 20 code metrics (D4) and seasonal terms (D2)
Hierarchical models, EB, profile likelihood	D2, D3, D4, D6, D8	Partial pooling across weeks (D2), systems (D3), releases (D4), collections (D6), band blocks (D8); HB versus EB; profile versus posterior intervals

4 Statistical framework (condensed)

Likelihood and exact posterior. For a uniform subset S of $\{1,\dots,N\}$ with $k=|S|$, $m=\max S$: $\mathbb{P}(S \mid N)=\binom{N}{k}^{-1}\mathbf{1}\{N\ge m\}$, so (m,k) is sufficient and the posterior $\pi(n \mid S)\propto\binom{n}{k}^{-1}\pi(n)\mathbf{1}\{n\ge m\}$ is one-dimensional and computable exactly. Under the flat prior,

$$\mathbb{P}(N\ge n \mid S)=\binom{m-1}{k-1}\Big/\binom{n-1}{k-1},\qquad \mathbb{E}[N \mid S]=\frac{(k-1)(m-1)}{k-2}\ (k\ge 3),$$

and the HPD interval is $[m,u_\alpha]$ with u_α the smallest n such that $\binom{m-1}{k-1}/\binom{n}{k-1}\le\alpha$.

Estimates and priors. Squared, absolute and 0–1 loss give the mean, median and MAP; asymmetric linear loss with cost ratio $c_u:c_o$ gives the $\tau=c_u/(c_u+c_o)$ quantile. Priors: flat, $1/n$ (scale-invariant; a formal Jeffreys-type choice since the regularity conditions fail for the uniform), truncated uniform on $\{1,\dots,\Omega\}$, Poisson(μ), and Gamma–Poisson (negative binomial marginal). In the continuum limit $U(0,\theta)$ the Pareto prior is conjugate, although the family is not an exponential family.

Bayes factors. $\mathrm{BF}_{10}=p(S \mid H_1)/p(S \mid H_0)$ with $p(S \mid H)=\sum_{n\ge m}\binom{n}{k}^{-1}\pi_H(n)$, which needs proper priors; for $H_0: N\le L$ with $m>L$ we get $p(S \mid H_0)=0$. Nested selection-bias tests use the Savage–Dickey ratio $\pi(\theta=0 \mid S)/\pi(\theta=0)$.

Thinning model (random k). $N\sim\mathrm{Pois}(\mu)$, each unit captured independently with probability p, $p\sim\mathrm{Beta}(a,b)$, $\mu\sim\mathrm{Gamma}(\alpha,\beta)$.

Proposition 1 (to be derived and checked numerically)**.** $N \mid S,\mu,p\sim\mathrm{Pois}(\mu(1-p))$ *truncated to* $n\ge m$; $p \mid N,S\sim\mathrm{Beta}(a+k,b+N-k)$; $\mu \mid N\sim\mathrm{Gamma}(\alpha+N,\beta+1)$. *Marginalising* p: $\pi(n \mid S,\mu)\propto\mathrm{Pois}(n;\mu)\mathrm{B}(a+k,b+n-k)$.

Selection bias. With weights $w(s;\theta)=s^{-\theta}$ and $k\ll N$, $\pi(N,\theta \mid S)\propto\pi(N)\pi(\theta)\prod_i w(s_i;\theta)/Z_N(\theta)^k\mathbf{1}\{N\ge m\}$, $Z_N(\theta)=\sum_{j\le N}w(j;\theta)$; the N-dependent normaliser must be kept. Sampler: exact discrete draw of N, random-walk MH for θ.

Gamma–Exponential (Jelinski–Moranda). Inter-failure times $t_i\sim\mathrm{Exp}(\phi(N-i+1))$, $\phi\sim\mathrm{Gamma}(a,b)$:

$$p(t_{1:k} \mid N)=\frac{b^a\Gamma(a+k)}{\Gamma(a)}\,\frac{N!/(N-k)!}{(b+T_N)^{a+k}},\qquad T_N=\sum_{i=1}^{k}(N-i+1)t_i,$$

so the posterior of N is exact on a grid.

Dirichlet–Multinomial and Normal–Gamma. Shares of F series: $\pi \mid \text{counts}\sim\mathrm{Dir}(\alpha+k_{1:F})$ when $k\ll N$ (exact likelihood: $\binom{\sum_f N_f}{k}^{-1}\prod_f\mathbf{1}\{N_f\ge m_f\}$). Log-rates $y_j\sim N(\eta,1/\tau)$ with a Normal–Gamma prior have closed-form posteriors and Student-t predictive.

Regression and hierarchy. With $y_j=\log\mu_j\sim N(x_j^\top\beta,\sigma^2)$: Normal-Inverse-Gamma posterior $V_J=(V_0^{-1}+X^\top X)^{-1}$, $\beta_J=V_JX^\top y$; ridge (Gaussian prior) and LASSO (Laplace prior) as MAP; spike-and-slab $\beta_l \mid \gamma_l\sim\gamma_lN(0,\sigma^2\tau^2)+(1-\gamma_l)\delta_0$, $\gamma_l\sim\mathrm{Bernoulli}(\omega)$, $\omega\sim\mathrm{Beta}$, sampled by collapsed Gibbs. Hierarchical Bayes puts hyperpriors on (η,σ^2,ω); empirical Bayes maximises the marginal likelihood; profile likelihood $L_P(N)=\max_\theta L(N,\theta)$ is compared with the posterior.

5 Analysis protocols by dataset

WP1 — Exact Bayesian estimation, priors and Bayes factors on invoice numbers (D1)

Data preparation. Load D1; separate cancellation invoices (leading "C"); take the distinct invoice numbers and form two label sets: (a) *dense labels*, the invoices ranked $1, \ldots, N_{\text{true}}$, which satisfy the textbook model exactly; (b) *raw labels*, the six-digit numbers themselves, which contain gaps. Record N_{true} and the gap structure.

Analysis.

1. For $k \in \{5, 10, 20, 50, 100\}$ draw $R = 2000$ random subsets of the dense labels. Compute the exact posterior under flat, $1/n$, truncated uniform and Poisson/negative binomial priors; for the informative priors set the prior mean at 0.5, 1 and 2 times a deliberately chosen guess to study sensitivity.
2. Report, over replicates: bias and RMSE of mean, median and MAP; the UMVUE and MLE as benchmarks; coverage and width of 95% HPD and equal-tailed intervals; log score; PIT/rank histograms.
3. **Bayes factors:** for $L \in \{0.5, 0.9, 1, 1.1\} \times N_{\text{true}}$ test $H_0 : N \leq L$ versus $H_1 : N > L$; plot BF against k; estimate false-alarm rate when $L \geq N_{\text{true}}$ and power when $L < N_{\text{true}}$; vary the prior width to display the Lindley–Jeffreys effect.
4. Repeat with raw labels to measure how gaps break coverage.

Success criterion. Dense labels: HPD coverage close to nominal; documented, quantified degradation for raw labels; a rule of thumb for the number of invoices needed for a given interval width.

WP2 — Thinning, selection bias and MCMC on recorded-customer invoices (D1)

Data. Natural sample: the set S of invoices that have a recorded customer, a thinning of the full invoice list with true capture fraction $\hat{p} = |S|/N_{\text{true}}$ known. Also random subsamples of S to mimic scarcity.

Analysis.

1. Check uniformity of S over the label range (Kolmogorov–Smirnov on s_i/m, spacing plots, observed minimum versus expected $\approx N/(k+1)$).
2. Fit the thinning model by the Gibbs sampler of Proposition 1, with several priors for (μ, p), and compare posteriors of N and p with N_{true} and $\hat{p}$.
3. Fit the selection-bias model by Metropolis–Hastings; Savage–Dickey Bayes factor for $\theta = 0$; profile likelihood for N.
4. Diagnostics: trace and autocorrelation plots, split-$\hat{R}$, ESS over four dispersed chains, posterior predictive check of (k, m). Compare MCMC output with the exact grid posterior in N (one dimension) and (N, θ) (two dimensions) by total variation.

Success criterion. MCMC reproduces the exact posterior (TV below an agreed tolerance); the report states whether recorded-customer invoices are uniform, with evidence.

WP3 — Production over time: Normal–Gamma, hierarchy, spike-and-slab (D2)

Data. Weekly invoice counts and cumulative counts N_t from D2, known exactly. At each weekend draw a random subset of invoices issued so far, giving a snapshot sample.

Analysis.

1. Estimate N_t at each snapshot and the weekly production $N_t - N_{t-1}$ as a *posterior difference* under the constraint $N_t \geq N_{t-1}$; compare with true weekly counts and with differences of point estimates.

2. Second stage: Normal–Gamma model for log weekly volume; posterior predictive interval for the next month, scored against the actual.
3. Regression of log volume on trend and seasonal terms (month indicators) with Normal-Inverse-Gamma, ridge/LASSO and spike-and-slab; posterior inclusion probabilities of seasonal effects.
4. Hierarchical Bayes versus empirical Bayes across weeks, with emphasis on weeks with few sampled invoices (shrinkage gain, interval widths).

Success criterion. Coverage of weekly-production intervals; lower error from partial pooling than from independent weekly estimates, quantified.

WP4 — Gamma–Exponential on real software failure data (D3)

Data. Inter-failure times for `sys1` (136 failures) and other DACS systems.

Analysis.

1. For the first $k_{\text{train}} \in \{30, 50, 80, 100\}$ failures, compute the exact Jelinski–Moranda posterior of N (and of the remaining defects $N - k$) under several priors on N and ϕ.
2. **Hold-out check:** compare predicted and observed numbers of failures in the remaining observation window; score the posterior predictive of the held-out inter-failure times by log score.
3. Posterior predictive checks of the fitted constant-rate model, and a comparison with a heterogeneous-rate extension.
4. Pool several systems hierarchically and compare EB with HB for the rate parameter.
5. Release rule: $\mathbb{P}(N - k \leq r \mid \text{data}) \geq 0.95$ and the asymmetric-loss quantile.

Success criterion. Honest assessment of when the model fits; calibrated predictive intervals for held-out failures.

WP5 — Bayesian linear models and hierarchical analysis on defect data (D4)

Data. Per-class bug counts and 20 code metrics (wmc, dit, noc, cbo, rfc, lcom, loc, and so on) for several releases; `hotspot` gives a reliable R route. Response $y = \log(1 + \text{bugs})$ with metrics log-transformed and standardised. These data have no serial labels, so they exercise the regression machinery; the fitted defect intensity supplies the prior mean for module-level defect counts in the thinning model.

Analysis.

1. Normal-Inverse-Gamma regression with closed-form posterior and marginal likelihood; Bayes factors between nested metric subsets.
2. Ridge and LASSO as MAP estimates; cross-validated penalty versus posterior; Bayesian lasso if time permits.
3. Spike-and-slab Gibbs sampler: posterior inclusion probabilities and posterior model probabilities, noting how correlated metrics (size, weighted methods, response for class) split inclusion probability.
4. Hierarchy over releases and projects: random intercepts, partial pooling, HB versus EB.
5. Temporal validation: train on release i, predict release $i + 1$ (RMSE, predictive log score).

Success criterion. Stable selected metrics across projects; better out-of-sample predictive score for hierarchical than pooled or separate fits.

WP6 — Hypothesis testing and selection bias on token IDs (D5, D6)

Data. CryptoPunk IDs $0, \ldots, 9999$ (shift by one), true supply 10,000. Samples: (i) random subsets for calibration; (ii) the unique IDs appearing in sales within the first w days of a window, a natural sample with k small. For D6, the same two sample types for each of eight collections with supplies read from the collection data.

Analysis.

1. Exact posteriors for random subsets (as in WP1) on a second real population.
2. Natural sample: uniformity tests; selection model for θ; compare estimates of N with 10,000 with and without selection.
3. Bayes factors for "supply $\leq L$" with L around the truth; false positives for $L = 10{,}000$.
4. Hierarchical model over the eight collections for θ_j and N_j; HB versus EB; coverage across collections.

Success criterion. Calibration across collections; a documented comparison of sales-based and random samples.

WP7 — Several series and counterfeit detection (D7 plus injected contamination)

Data preparation. Download the registry; keep purely numeric serials; group by make and model code via the reference table; choose a few families with dense numeric serial ranges. Serials are free text, so cleaning rules are logged.

Analysis.

1. Dirichlet–Multinomial posterior for shares of registered aircraft by model within a manufacturer; exact multi-series posterior for (N_f).
2. Compare posteriors with published production totals for chosen families (sourced and cited by the student). Survivorship makes the register a non-uniform sample, so also fit the selection model.
3. **Semi-synthetic counterfeit experiment:** inject cloned (duplicate) serials and out-of-range serials at known rates $\varepsilon \in \{0, 1, 5, 10\}\%$ into a real series (from D7 or D1). Fit the Beta–Bernoulli contamination model by Gibbs sampling; report posterior of ε, posterior odds that N exceeds a legal cap, and ROC of the posterior odds as a detector against the rule "maximum serial exceeds the cap".

Success criterion. Recovery of ε with calibrated intervals; documented advantage (or not) over the simple rule.

WP8 — Two-stage wildlife model (D8, D9)

Stage 1 (tags deployed, D8). Choose one species group and a band block (prefix series); the number of banding records in the block gives a ground truth for the number of bands used. Samples: random subsets of band numbers; and band numbers of birds that have encounter records, a natural, non-uniform sample (for example, harvested birds). Estimate the number of bands used, fit the selection model, and compare across several blocks with HB versus EB.

Stage 2 (population, D9). From the `hare` data take occasion 1 as marking (M animals), occasion 2 as recapture sample (n animals, r marked). Beta–Binomial posterior $p \mid r \sim \text{Beta}(a + r, b + n - r)$ with $N_{\text{pop}} = M/p$; compare with the Chapman estimator and with the published log-linear estimates of about 77 to 81. Compose stages 1 and 2 by sampling to show how uncertainty in M propagates.

Success criterion. Posterior for abundance consistent with published estimates; clear statement of what ring numbers can and cannot identify.

Historical benchmark (WWII table)

The aggregate table (statistical estimates $169,244,327$; German records $122,271,342$) is reproduced as a figure and used to motivate priors. No serial-number data are public, so no model is fitted to it.

6 Evaluation protocol

- **Design.** Every dataset with ground truth is analysed over $R \geq 1000$ replicate subsamples at several k; random seeds fixed and logged.
- **Metrics.** Bias, RMSE, log score, CRPS, interval score, coverage and width of 95% credible intervals, rank (SBC) histograms, and for samplers ESS per second and total variation from the exact posterior.
- **Benchmarks.** UMVUE, MLE, frequentist intervals, Chapman estimator, LASSO and ridge, EB versus HB.
- **Assumption diagnostics as data analysis.** For every natural sample: uniformity tests, spacing plots, minimum check, and a comparison of results with and without a selection model. Negative results are reported.

7 Data management, reproducibility and ethics

- A data dictionary and cleaning log for each dataset; raw files are not committed, only download scripts with checksums; analysis scripts and notebooks under version control (R and Python).
- Licences: D1 and D2 are CC BY 4.0 (cite the donor); D8 is CC0; D7 is a public government record; D9 and D3 ship with CRAN packages. Check the terms of D4–D6 before redistributing anything.
- All data are public. `CustomerID` values and blockchain addresses are pseudonymous; they are not used to profile individuals, and no attempt is made to identify them.
- The counterfeit study uses injected, labelled contamination and makes no claims about real counterfeiters.

8 Timeline (12 weeks)

Weeks	Tasks and data milestones
1	Download D1, D2, D3, D4; data dictionary and cleaning log; check label structure, gaps and uniformity; baseline exact-posterior code
2	WP1: estimation, priors, HPD, coverage on dense invoice labels
3	WP1: Bayes factors, prior sensitivity, raw labels versus dense labels
4–5	WP2: thinning Gibbs sampler, selection MH sampler, diagnostics, comparison with exact grids
6	WP4: Jelinski–Moranda on D3 with hold-out validation
7	WP3: weekly production, Normal–Gamma, hierarchy (D2)
8	WP5: regression, ridge/LASSO, spike-and-slab, hierarchy on D4
9	WP6: CryptoPunks and eight-collection analysis (D5, D6)
10	WP7: multi-series on D7 and counterfeit injection study
11	WP8: bird-banding and `hare` analysis (D8, D9), if time permits; otherwise lighter version
12	Final report, figures, code clean-up, presentation

Priority. Core: WP1, WP2, WP3, WP4, WP5, WP6. Secondary: WP7 and WP8 (a reduced version of each is acceptable).

9 Risks

- **Violated assumptions:** gaps in invoice numbers, non-uniform natural samples and free-text serials. This is itself a finding; protocols include diagnostics and report both results.
- **Data access and size:** bird-banding files are large; use one zipped group or a Bander Portal extract. The original PROMISE host may be offline, so use the `hotspot` package or a mirror. Kaggle and the Cornell data need an account or manual download.
- **Ground truth:** verify the supply of every NFT collection and the production totals for aircraft families against a primary source before use.
- **Scope:** priority order above; the neural posterior estimator from earlier versions is dropped to leave room for data analysis.

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